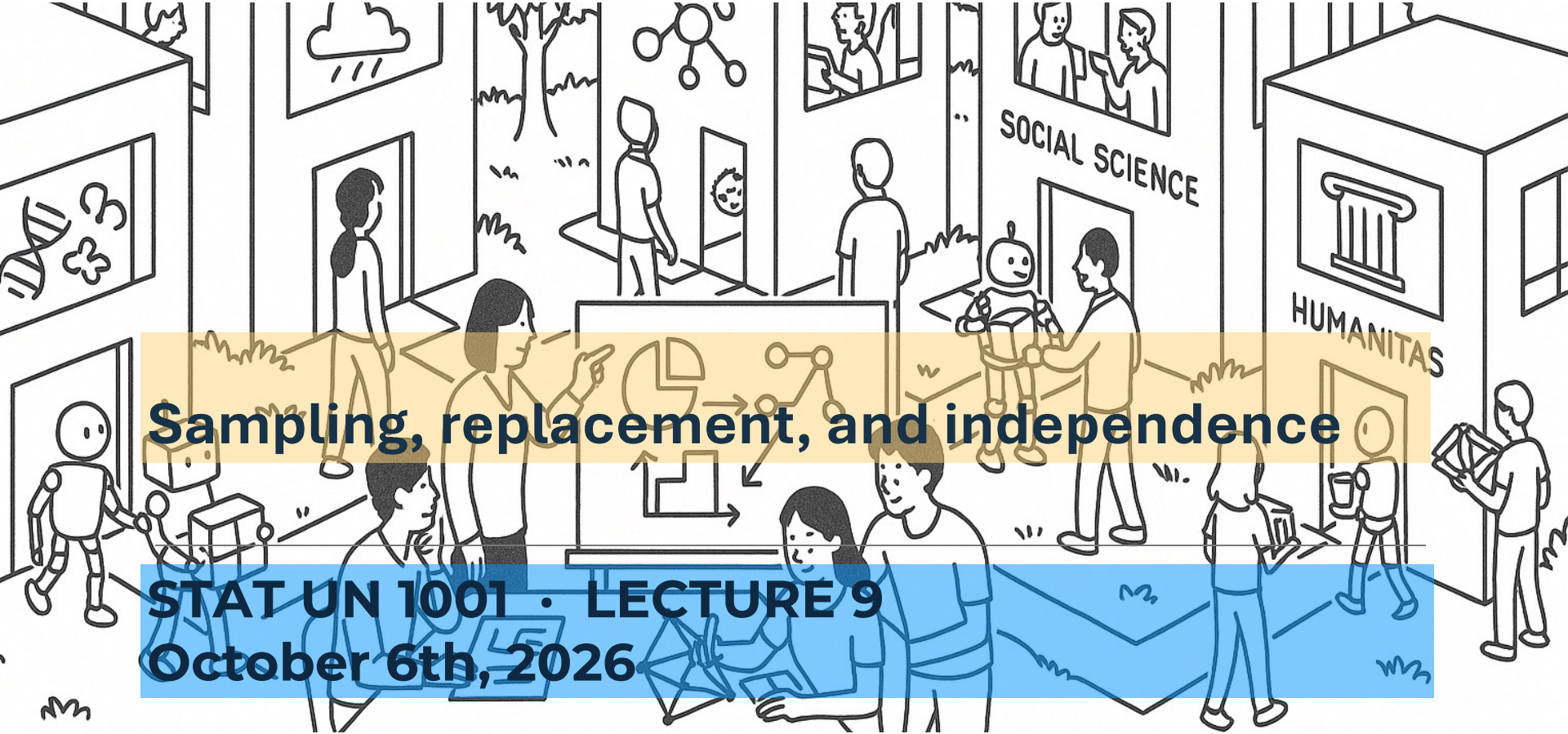


Sampling, replacement, and independence

STAT UN 1001 · LECTURE 9

October 6th, 2026

The five-part setup

1	**Chance process**	What random action repeats? Is it the same process?
2	**Sample space**	What outcomes could happen?
3	**Event**	Which outcomes count?
4	**Probability sentence**	What probability are we finding?
5	**Rule**	Complement, multiplication, table, or tree?

Small population example

A study group has five practice cards. Two cards are probability, one is graphs, one is study design, and one is numerical summaries.

Card	A	B	C	D	E
Topic	Prob.	Prob.	Graphs	Study design	Summaries

QUESTION

If you draw two cards, what changes after the first draw?

The answer depends on replacement.

With replacement

WHAT HAPPENS

Draw a card, record its topic, **put it back**, mix again, and draw the second card.

PROBABILITY CONSEQUENCE

The second draw still has five possible cards. The chance of the *probability* card stays 2/5.

INDEPENDENCE CONNECTION

The first draw does not change the probability for the second draw.

Everyday version: a playlist that allows repeats can play the same song again.

Without replacement

WHAT HAPPENS

Draw a card, **keep it out**, then draw the second card from the four cards that remain.

PROBABILITY CONSEQUENCE

The denominator changes from 5 to 4. The numerator may change too.

INDEPENDENCE CONNECTION

The first draw can change the probability for the second draw.

Everyday version: a no-repeat playlist changes what can come next.

Two second-draw probabilities

Sampling method	First draw	Cards left	**P(second card is probability)**
Without replacement	***Probability***	1 probability card among 4 cards	1/4
	Not ***probability***	2 probability cards among 4 cards	2/4
With replacement	*Any card*	2 probability cards among 5 cards	2/5

MAIN INSIGHT **Independence asks whether knowing the first outcome changes the probability of the next event.**

The same question can have different answers under different sampling rules.

Small versus large populations

Population	Remove one person	Effect on next draw
15 students	1/15 of the population	Large enough to notice
1,500 students	1/1,500 of the population	Usually tiny

PRACTICAL READING

Without replacement is not independent, but it can behave almost like independence when the population is huge compared with the sample.

This is why survey sampling often cares about sample size relative to population size.

Dinner-party seating

Eight people sit randomly around a round table. Your seat is fixed. You care whether Alex sits next to you.

CHANCE PROCESS

Randomly assign the seven other people to the seven remaining seats.

SAMPLE SPACE

There are 7 possible seats for Alex, not 8, because your seat is already occupied.

PROBABILITY

Two seats are next to you, so P(Alex sits next to you) = 2/7.

Raffle rules

A raffle has 30 tickets and 7 prizes. You bought one ticket.

Rule	What repeats?	Your chance of at least one prize
Without replacement	Seven different winning tickets	7/30
With replacement	Your ticket could be drawn on any prize draw	1 - (29/30)^7

A raffle question is incomplete until we know the replacement rule.

Socks and complements

Drawer: 4 blue socks, 5 gray socks, 3 black socks. You grab 2 socks without replacement.

MATCHING SOCKS

The event includes blue-blue, gray-gray, or black-black.

AT LEAST ONE BLACK SOCK

The complement is no black socks.

TWO BLUE SOCKS

The probability sentence is (4/12)(3/11). The second fraction changes because one blue sock has been removed.

Independent is not the same as disjoint

Concept	Question it asks	Everyday wording
Disjoint	Can both happen at once?	One lunch order cannot be both only soup and only salad
Overlapping	Can both happen at once?	A student can have an iPhone and an Apple computer
Independent	Does knowing one change the probability of the other?	A replacement rule can make repeated draws independent

Disjoint is about possibility. Independent is about information.

In-class practice

For each situation, decide: with replacement, without replacement, or neither clear from the wording. Then say whether independence is reasonable.

1. A music app shuffles five songs and allows repeats.

2. A TA draws three student names from a hat and does not put names back.

3. A website shows survey respondents one of two versions, but the assignment algorithm is not described.

Good habits for working out problems

BEFORE CALCULATING

Name the chance process, sample space, event, and denominator.

BEFORE MULTIPLYING

Ask whether the next stage uses the same population or a changed population.

BEFORE ASSUMING INDEPENDENCE

Ask whether knowing one outcome changes the probability of another.

Takeaway

Probability is not just calculation. It is a way to describe how a random process unfolds and repeats.

WITH REPLACEMENT

The process resets.

WITHOUT REPLACEMENT

The process remembers what has already happened.

INDEPENDENCE

Knowing one event does not change the probability of the other.